SQ-126

Seat No.	

Total No. of Pages : 2

B. Tech. (Part - IV) (Semester -VII)
Examination, May-2025
ELECTRICAL
Switchgear and protection
Sub. Code : 83779/67517/83974

Day and Date : Tuesday, 06/05/2025 **Total Marks : 70**

Time : 10:30 a.m. to 01:00 p.m.

Instructions : 1) All questions are compulsory.
2) Use non-programmable calculator is permissible.
3) Figures to the right indicate full marks.
4) Assume suitable data if required.

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SECTION-I

Q1) A) Explain Air break circuit breaker in detail. **[6]**

B) Write note on : **[6]**

i) Arc Voltage

ii) Transient Restriking voltage.

OR

B) Explain principles of DC and AC are interruption in detail.

Q2) A) Write note on induction cup type relay in detail. **[6]**

B) Explain microprocessor based over voltage relay in detail. **[6]**

OR

B) Which are the desirable qualities of protective relays? Explain in detail.

P.T.O.

Q3) A) Explain directional relay in detail. **[6]**

B) Explain Ring main protection scheme in detail. **[6]**

OR

B) Write note on Plug setting in detail.

SECTION-II

Q4) A) Draw and explain block schematic of carrier aided protection. **[6]**

B) Explain working of percentage differential relay. **[6]**

OR

B) Explain carrier current protection scheme for line protection.

Q5) A) Draw a neat sketch of Buchholz relay and explain its working principle. State its application. **[6]**

B) Explain problems associated with percentage differential protection. **[6]**

OR

B) Explain the restricted earth fault protection for transformer.

Q6) A) Explain excitation failure and protection against motoring in generator. **[5]**

B) Explain in detail microprocessor based mho relay. **[5]**

OR

B) Explain relay setting for 3 zone distance protection.

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